

Fire-fighting System

HyaDuo 2 D FL

Type Series Booklet



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Type Series Booklet HyaDuo 2 D FL

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Building Services: Water Supply

Fire-fighting Systems

HyaDuo 2 D FL



Main applications

- Fire-fighting systems to DIN 14462

Fluids handled

- Fire-fighting water
- Liquids not containing aggressive, abrasive or solid substances

Operating data

Table 1: Operating properties

Characteristic		Value
Flow rate	Q [m³/h]	≤ 110
	Q [l/s]	≤ 30,5
Head	H [m]	≤ 160
Fluid temperature	T [°C]	≥ 0
		≤ +70
Operating pressure	p _d [bar]	≤ 16
Inlet pressure	p _{inl} [bar]	≤ 6
Permissible inlet pressure fluctuation	p [bar]	+0,3/-0,2
Power supply	U [V]	3/PE, AC 400, 50 Hz

Design details

Design

- Fully automatic fire-fighting system
- Baseplate-mounted
- 2 discharge-side, direct-flow membrane-type accumulators, approved for drinking water
- Vibration damping

Installation

- Stationary dry installation

Drive

- Three-phase asynchronous motor
- 220 – 240 V / 380 – 420 V, 380 – 420 V / 660 – 720 V
- Enclosure IP55
- Efficiency class IE3 to IEC 60034-30
- DOL starting up to and including 4 kW
- Soft starter from and including 5.5 kW

Automation

- Electro-mechanical control system, IP54
- 2 manual-automatic selector switches
- 2 external master switches
- 2 timers for functional check run every 24 hours and hygienic flushing function of the inlet line once a week
- 2 phase monitoring relays
- In automatic mode, the motor protection circuit only outputs an alert.
- Terminal strip/terminals with identification for all connections
- 2 pressure switches on the discharge side for starting and stopping, monitored for broken wires and short circuit
- 2 pressure switches for monitoring pressure build-up
- Pressure-controlled starting and pressure-controlled stopping
- 2 solenoid valves for temperature-dependent opening of the bypass line in the event of pumps overheating
- Control system with integrated flushing function and additional connection for optional flushing valve
- Circuit diagram to VDE and parts list for electric parts

Messages via signal lamps:

- Colour signal lamp indicating lack of water and no pressure build-up (red)
- Colour signal lamp indicating a fault (amber)
- Colour signal lamp indicating readiness for operation (green)
- Colour signal lamp indicating operation started by pressure switch or remote ON/OFF (white)

Indication via volt-free isolating terminals:

- Lack of water
- No pressure build-up
- Switch position of manual-automatic selector switch
- Pump control via pressure switch or remote ON/OFF
- General fault message
- Drinking water stop

Monitoring for broken wires and monitoring for short circuit:

- > 10.3 kΩ: broken wire
- 10 kΩ: good state
- 1 kΩ: pump start through actuation of limit switch
- < 300 Ω: short circuit

Designation

Example: HyaDuo 2 D FL 2 / 1506

Table 2: Designation key

Code	Description
HyaDuo	Type series
2	Generation
D FL	Fire-fighting system with three-phase motor
2	Number of pump sets
15	Pump size
06	Number of stages

Configuration and function

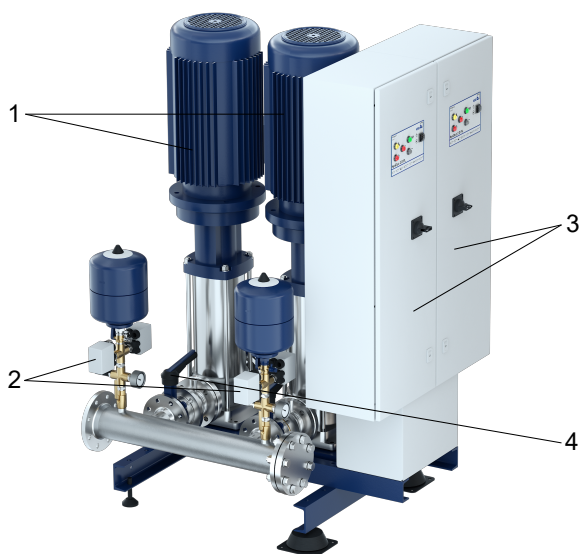


Fig. 1: Illustration of the fire-fighting system

1	High-pressure centrifugal pumps	3	Control cabinets
2	Pressure switch	4	Valve

Design

Fully automatic fire-fighting system on anti-vibration mounts, comprising a dual-pump system with redundant function, piping and control cabinets (3) on a common baseplate. The fire-fighting system comprises one duty system and one back-up system.

Function

The pump sets are started and stopped as a function of pressure via an electro-mechanical control system. If the pressure drops below the set start-up pressure p_E the pump sets are started up via the pressure switch. When consumption decreases again, the pump sets are stopped as a function of pressure after the set minimum runtime (up to 10 minutes). Independently of the pressure switch, the fire-fighting system can also be controlled via remote ON/OFF (e.g. a switch in the wall hydrant). The pressure switch and remote ON/OFF input are monitored for broken wires and short circuit. If a broken wire or short circuit are detected, the fire-fighting system is started up automatically and a fault is indicated.

Dry running protection stops the pump sets in automatic mode as soon as no more fluid is pumped (default setting). Tripping of the motor protection switch triggers a fault message.

Materials

Table 3: Overview of available materials

Part No.	Description	Material
Pump		
101	Inlet casing	Stainless steel
101	Discharge casing	Stainless steel
655	Hydraulic system	Stainless steel
411	Sealing element	FPM 80
-	Plain bearing	Aluminium oxide
433	Mechanical seal	To EN 12756
433.01	Primary ring	Silicon carbide
433.02	Mating ring	Hard carbon
-	Elastomer	EPDM
Hydraulic design		
-	Piping	Chrome steel
79.01	Control unit	Steel, painted
-	Accumulator	Connection made of stainless steel
-	Membrane	Approved for drinking water
-	Valves	Movitec 2B to 15C: copper-base alloy / brass Movitec 25B to 90B: GG/EPDM

Product benefits

- Design to DIN 14462 ensures reliable operation in the event of fire.
- High operational availability by regular functional check run
- Reliable operation by automatic minimum flow system
- Reliable operation with pressure switch and remote ON/OFF monitored for broken wires and short circuit
- Operating reliability by redundant system design
- Hygienically safe with integrated timer controlling the flushing device in the drinking water inlet line
- Anti-vibration mounts prevent solid-borne sound transmission.

Product information

Product information as per Regulation No. 1907/2006 (REACH)

For information as per European chemicals regulation (EC) No. 1907/2006 (REACH) see <https://www.ksb.com/en-global/company/corporate-responsibility/reach>.

Selection information

Lightning protection

- Electrical installations must be protected against overvoltage (binding since 14 December 2018) (see DIN VDE 0100-443 (IEC 60364-4-44:2007/A1:2015, modified) and DIN VDE 0100-534 (IEC 60364-5-53:2001/A2:2015, modified)). Whenever modifications are made to existing installations, retrofitting a surge protective device (SPD) in accordance with VDE is mandatory.
- A maximum cable length of 10 metres should not be exceeded between the surge protective device (usually type 1, internal lightning protection) installed at the service entrance and the equipment to be protected. For longer cables, additional surge protective devices (type 2) must be provided in the sub-distribution board upstream of the equipment to be protected or directly in the equipment itself.
- The associated lightning protection concept must be provided by the operator or by a suitable provider commissioned by the operator. Surge protective devices can be offered for the control units on request.

Selecting the fire-fighting system

 For selecting the fire-fighting system see KSB EasySelect and/or the Planning Information for Fire-fighting Water Systems (reference number 2300.023).

Example

Specify the operating points:

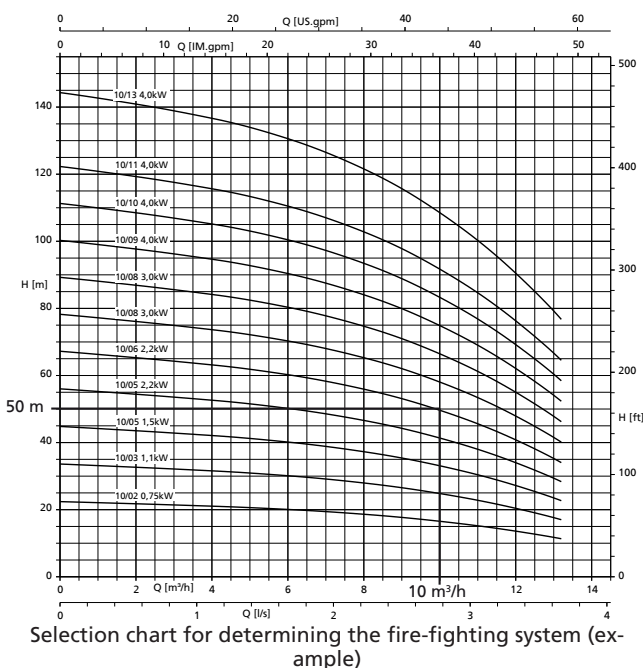
- Flow rate required: 10 m³/h
- Head: 50 m (start-up pressure p_g : 5.0 bar)

Calculation:

- Transfer the values to the selection chart to select the corresponding pump size.

⇒ Movitec 10/06

⇒ HyaDuo 2 D FL 2 / 1006



Technical data

HyaDuo 2 D FL with Movitec V

Table 4: Technical data

Size	Conne- ction	Nominal power	Nominal cur- rent	Starting cur- rent (soft start) ¹⁾	Minimum starting cur- rent	Total rated power	Mat. No.	[kg]
		P _N	I _N					
		[kW]	3~ 400 V, 50 Hz [A]					
2 / 0202	R 2	0,37	0,94	-	-	0,70	29135376	123
2 / 0203	R 2	0,37	0,94	-	-	0,70	29135377	124
2 / 0204	R 2	0,37	0,94	-	-	0,70	29135378	125
2 / 0205	R 2	0,37	0,94	-	-	0,70	29135379	126
2 / 0206	R 2	0,55	1,33	-	-	0,90	29135380	130
2 / 0207	R 2	0,55	1,33	-	-	0,90	29135381	131
2 / 0208	R 2	0,55	1,33	-	-	0,90	29135382	132
2 / 0209	R 2	0,75	1,68	-	-	1,20	29135383	136
2 / 0210	R 2	0,75	1,68	-	-	1,20	29135384	137
2 / 0211	R 2	1,10	2,40	16,80	6,70	1,70	29135385	143
2 / 0212	R 2	1,10	2,40	16,80	6,40	1,70	29135386	144
2 / 0214	R 2	1,10	2,40	16,80	6,70	1,70	29135387	145
2 / 0216	R 2	1,50	2,92	22,50	9,00	2,00	29135388	154
2 / 0218	R 2	1,50	2,92	22,50	9,00	2,00	29135389	156
2 / 0402	R 2	0,37	0,94	-	-	0,70	29135390	123
2 / 0403	R 2	0,55	1,33	-	-	0,90	29135391	128
2 / 0404	R 2	0,55	1,33	-	-	0,90	29135392	129
2 / 0405	R 2	0,75	1,68	-	-	1,20	29135393	133
2 / 0406	R 2	1,10	2,40	16,80	6,70	1,70	29135394	139
2 / 0407	R 2	1,10	2,40	16,80	6,70	1,70	29135395	140
2 / 0408	R 2	1,50	2,92	22,50	9,00	2,00	29135396	147
2 / 0409	R 2	1,50	2,92	22,50	9,00	2,00	29135397	149
2 / 0410	R 2	1,50	2,92	22,50	9,00	2,00	29135398	150
2 / 0411	R 2	2,20	4,15	39,60	12,80	2,90	29135399	156
2 / 0412	R 2	2,20	4,15	39,60	12,80	2,90	29135400	157
2 / 0414	R 2	2,20	4,15	39,60	12,80	2,90	29135401	159
2 / 0416	R 2	3,00	5,59	23,40	19,70	3,90	29135402	187
2 / 0602	R 2	0,37	0,94	-	-	0,70	29135403	125
2 / 0603	R 2	0,75	1,68	-	-	1,20	29135404	134
2 / 0604	R 2	1,10	2,40	16,80	6,70	1,70	29135405	139
2 / 0605	R 2	1,10	2,40	16,80	6,70	1,70	29135406	140
2 / 0606	R 2	1,50	2,92	22,50	9,00	2,00	29135407	149
2 / 0607	R 2	1,50	2,92	22,50	9,00	2,00	29135408	150
2 / 0608	R 2	2,20	4,15	39,60	12,80	2,90	29135409	157
2 / 0609	R 2	2,20	4,15	39,60	12,80	2,90	29135410	158
2 / 0610	R 2	2,20	4,15	39,60	12,80	2,90	29135411	159
2 / 0611	R 2	3,00	5,59	23,40	19,70	3,90	29135412	178
2 / 0612	R 2	3,00	5,59	23,40	19,70	3,90	29135413	179
2 / 0614	R 2	3,00	5,59	23,40	19,70	3,90	29135414	180
2 / 0616	R 2	4,00	7,45	36,70	25,30	5,20	29135415	201
2 / 1002	R 2	0,75	1,68	-	-	1,20	29135416	158
2 / 1003	R 2	1,10	2,40	16,80	6,70	1,70	29135417	164
2 / 1004	R 2	1,50	2,92	22,50	9,00	2,00	29135418	175
2 / 1005	R 2	2,20	4,15	39,60	12,80	2,90	29135419	182
2 / 1006	R 2	2,20	4,15	39,60	12,80	2,90	29135420	184
2 / 1007	R 2	3,00	5,59	23,40	19,70	3,90	29135421	202
2 / 1008	R 2	3,00	5,59	23,40	19,70	3,90	29135422	203
2 / 1009	R 2	4,00	7,45	36,70	25,30	5,20	29135423	216

¹⁾ Recommended triggering characteristic of fuse elements: gG or C

Size	Conne- ction	Nominal power	Nominal cur- rent	Starting cur- rent (soft start) ¹⁾	Minimum starting cur- rent	Total rated power	Mat. No.	[kg]
		P _N	I _N					
			3~ 400 V, 50 Hz					
		[kW]	[A]	[A]	[A]	[kVA]		
2 / 1010	R 2	4,00	7,45	36,70	25,30	5,20	29135424	218
2 / 1011	R 2	4,00	7,45	36,70	25,30	5,20	29135425	220
2 / 1013	R 2	5,50	10,00	51,00	35,20	6,90	29135426	327
2 / 1502	DN 80	2,20	4,15	39,60	12,80	2,90	29135427	204
2 / 1503	DN 80	3,00	5,59	41,90	19,70	3,90	29135428	225
2 / 1504	DN 80	4,00	7,45	36,70	25,30	5,20	29135429	235
2 / 1505	DN 80	5,50	10,00	51,00	35,20	6,90	29135430	339
2 / 1506	DN 80	7,50	13,40	68,10	47,20	9,30	29135431	341
2 / 1507	DN 80	7,50	13,40	68,10	47,20	9,30	29135432	351
2 / 1508	DN 80	11,00	19,30	89,60	61,80	13,40	29135433	412
2 / 2502	DN 100	4,00	7,45	36,70	25,30	5,20	29135434	381
2 / 2503	DN 100	5,50	10,00	51,00	35,20	6,90	29135435	464
2 / 2504	DN 100	7,50	13,40	68,10	47,20	9,30	29135436	478
2 / 2505	DN 100	11,00	19,30	89,60	61,80	13,40	29135437	657
2 / 2506	DN 100	11,00	19,30	89,60	61,80	13,40	29135438	663
2 / 2507	DN 100	15,00	26,20	121,60	83,80	18,20	29135439	692
2 / 4002-2	DN 100	5,50	10,00	51,00	35,20	6,90	29135440	420
2 / 4002	DN 100	7,50	13,40	68,10	47,20	9,30	29135441	428
2 / 4003-2	DN 100	11,00	19,30	89,60	61,80	13,40	29135442	585
2 / 4003	DN 100	11,00	19,30	89,60	61,80	13,40	29135443	586
2 / 4004-2	DN 100	15,00	26,20	121,60	83,80	18,20	29135444	595
2 / 4004	DN 100	15,00	26,20	121,60	83,80	18,20	29135445	623
2 / 4005-2	DN 100	18,50	31,80	147,20	101,80	22,00	29135446	533
2 / 4005	DN 100	18,50	31,80	147,20	101,80	22,00	29135447	563
2 / 4006-2	DN 100	18,50	31,80	147,20	101,80	22,00	29135448	568
2 / 4006	DN 100	22,00	37,60	185,30	127,80	26,10	29135449	640
2 / 6001	DN 150	5,50	10,00	51,00	35,20	6,90	29135450	490
2 / 6002-2	DN 150	7,50	13,40	68,10	47,20	9,30	29135451	505
2 / 6002	DN 150	11,00	19,30	89,60	61,80	13,40	29135452	659
2 / 6003-2	DN 150	15,00	26,20	121,60	83,80	18,20	29135453	699
2 / 6003	DN 150	18,50	31,80	147,20	101,80	22,00	29135454	706
2 / 6004-2	DN 150	18,50	31,80	147,20	101,80	22,00	29135455	647
2 / 6004	DN 150	22,00	37,60	185,30	127,80	26,10	29135456	726
2 / 6005-2	DN 150	22,00	37,60	185,30	127,80	26,10	29135457	733
2 / 9002-2	DN 150	11,00	19,30	89,60	61,80	13,40	29135458	780
2 / 9002-1	DN 150	15,00	26,20	121,60	83,80	18,20	29135459	797
2 / 9002	DN 150	15,00	26,20	121,60	83,80	18,20	29135460	797
2 / 9003-2	DN 150	18,50	31,80	147,20	101,80	22,00	29135461	756
2 / 9003-1	DN 150	22,00	37,60	185,30	127,80	26,10	29135462	828
2 / 9003	DN 150	22,00	37,60	185,30	127,80	26,10	29135463	828

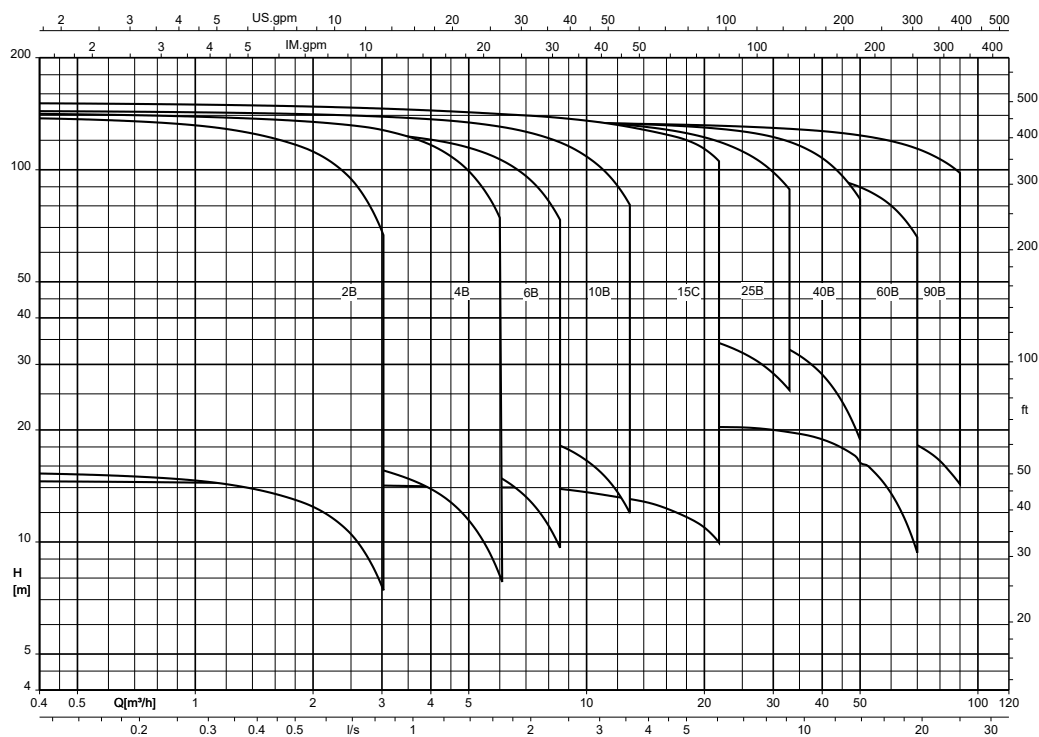
Selection aid for control cabinets

Table 5: Overview of control cabinets

Motor rating per pump	Maximum connection cross-section		Size of each control cabinet H x W x D
	Stranded	Solid	
[kW]	[mm ²]	[mm ²]	[mm]
5,5	6	10	1000 x 400 x 300
7,5	6	10	1000 x 400 x 300
11	10	16	1000 x 400 x 300
15	16	25	1000 x 400 x 300
18,5	16	25	1000 x 400 x 300
22	16	25	1000 x 400 x 300

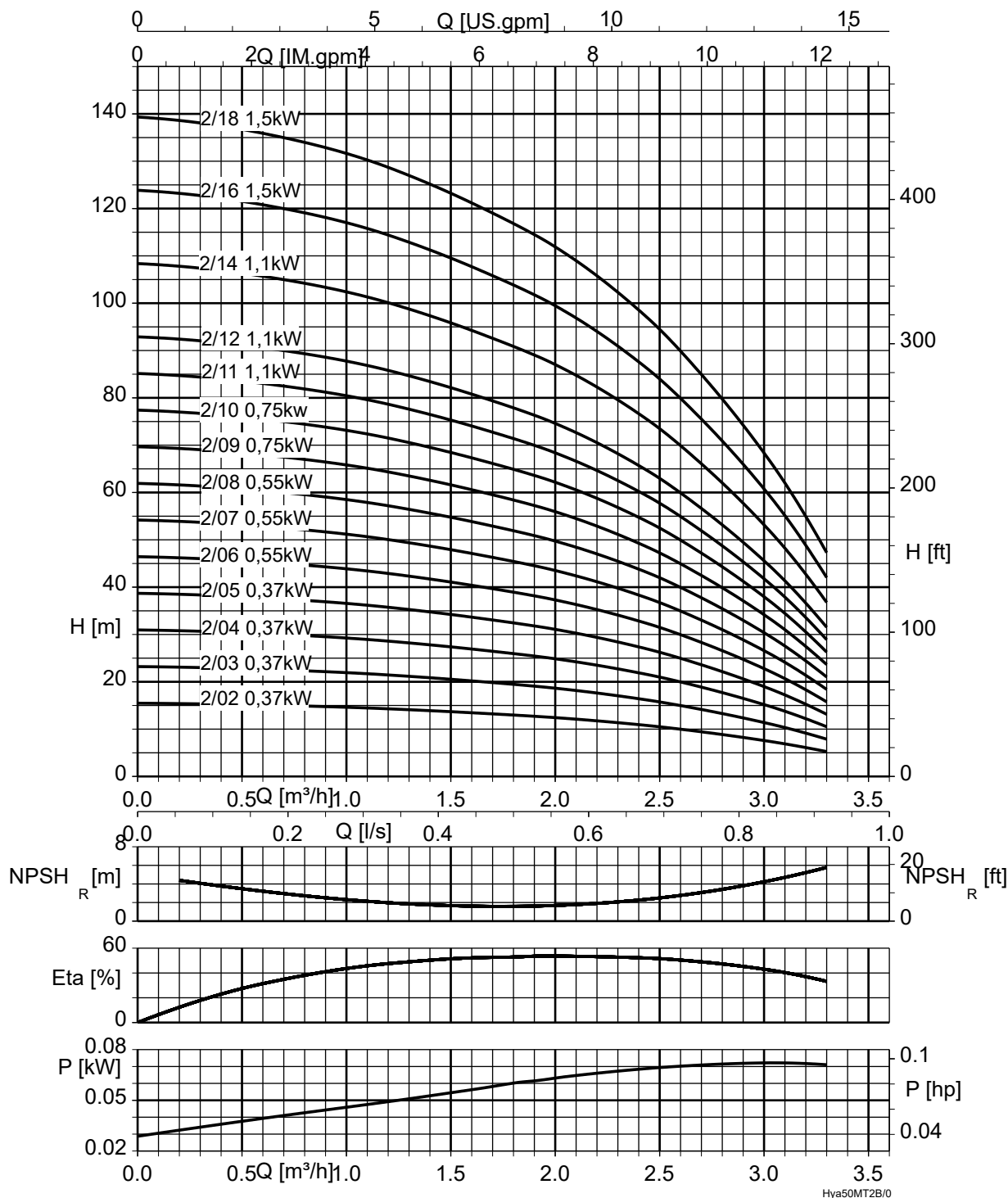
Selection chart

HyaDuo 2 D FL, n = 2900 rpm



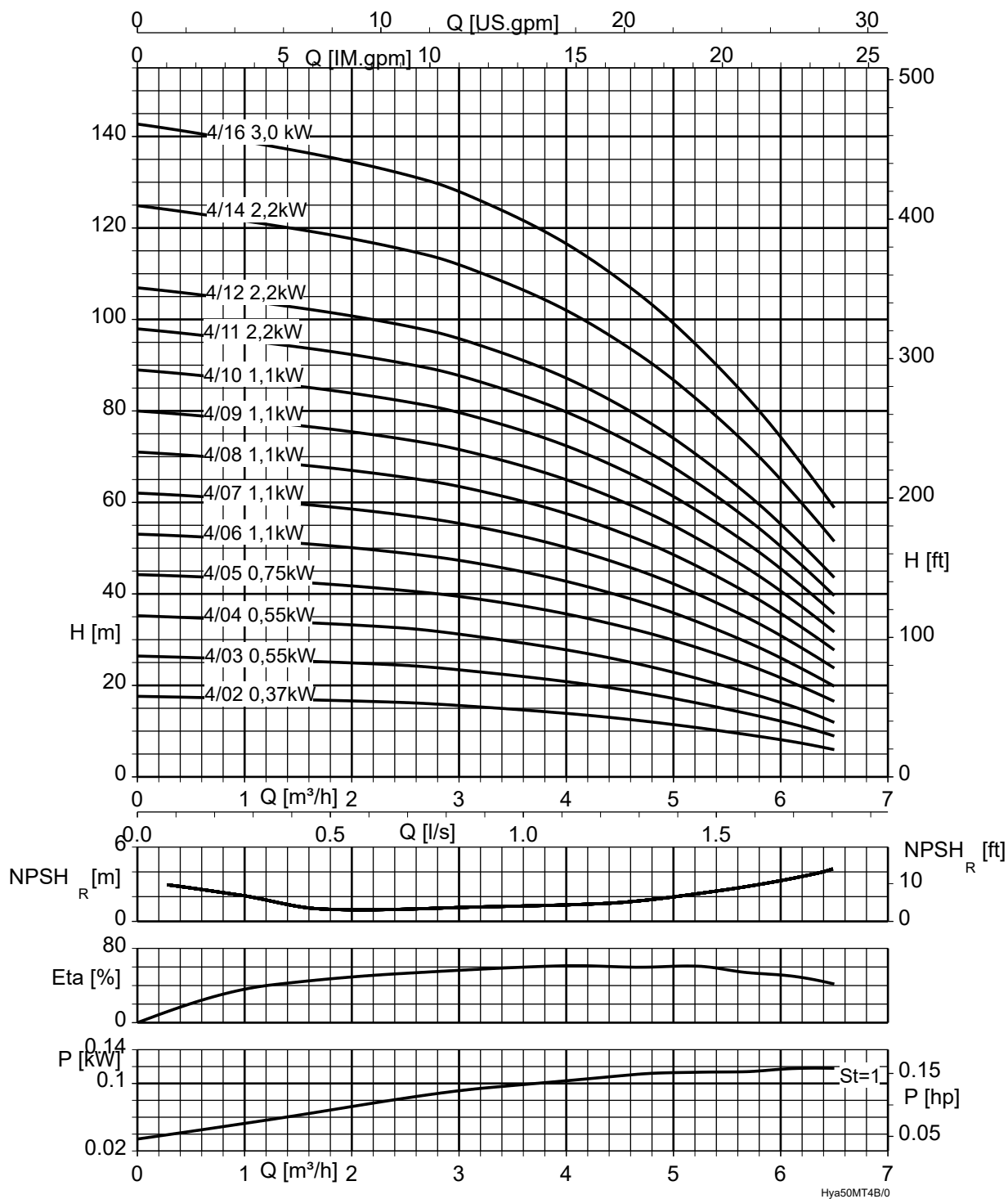
Characteristic curves

HyaDuo 2 D FL with Movitec 2B, n = 2900 rpm



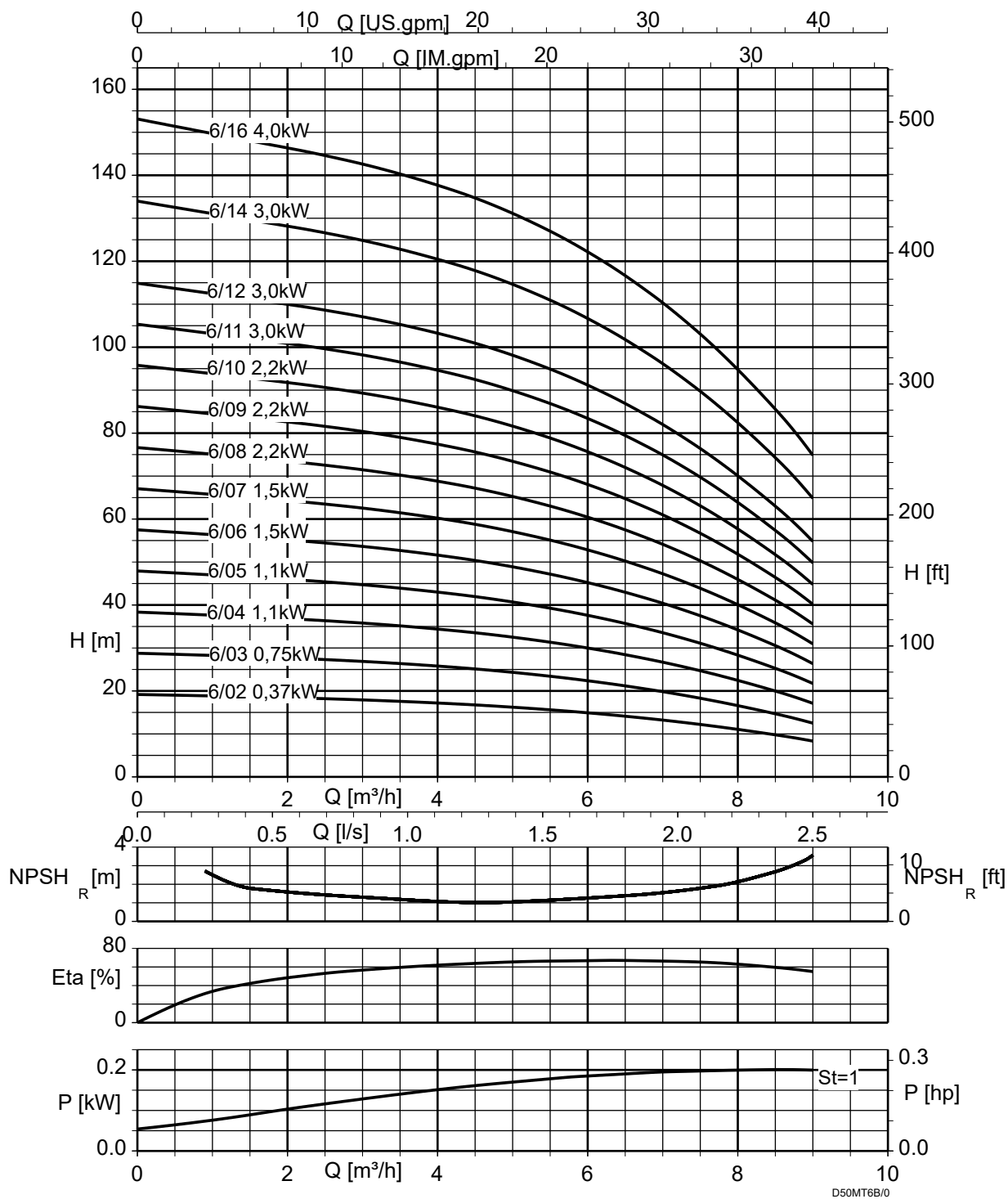
St = 1 P per stage

HyaDuo 2 D FL with Movitec 4B, n = 2900 rpm



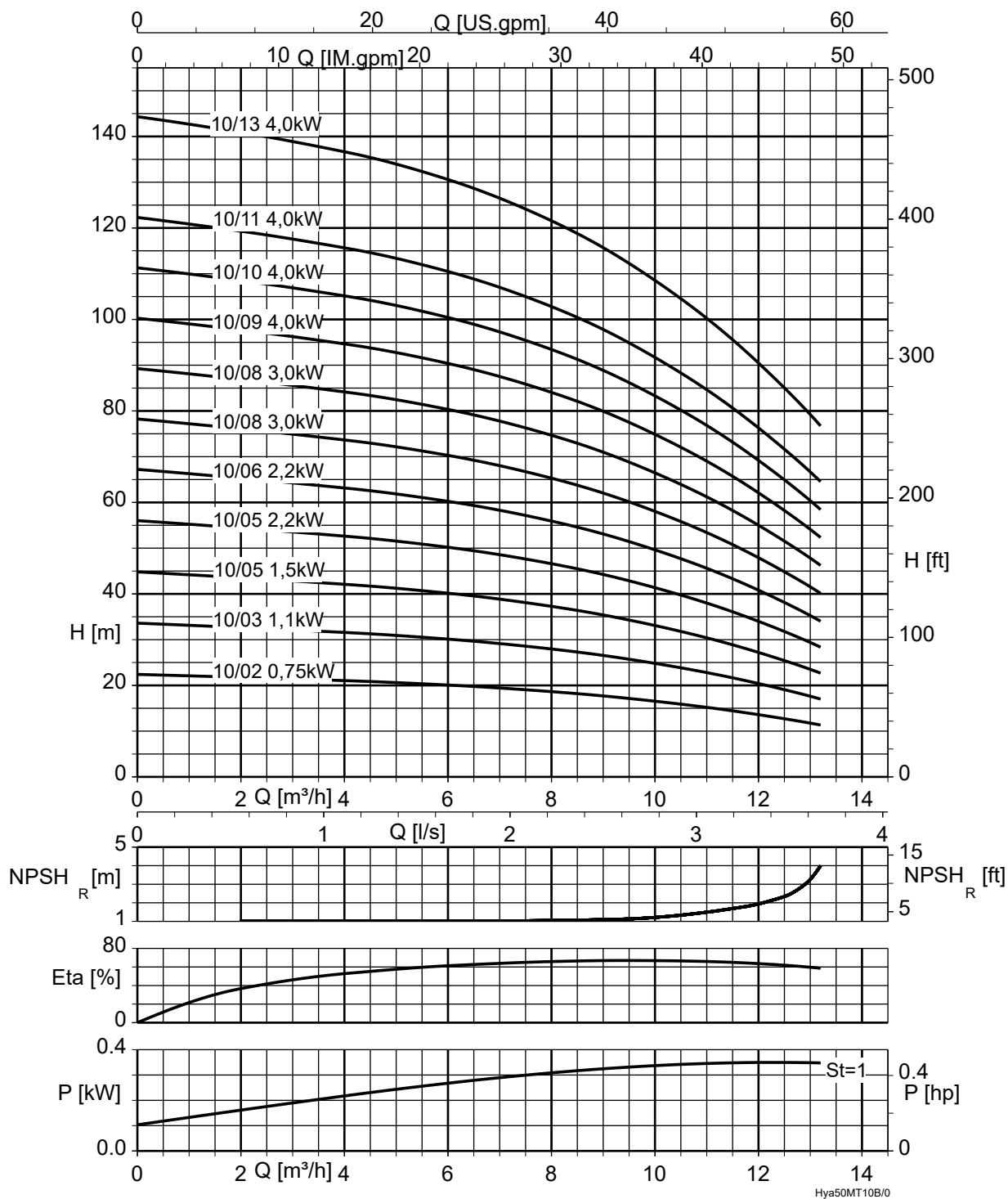
St = 1 | P per stage

HyaDuo 2 D FL with Movitec 6B, n = 2900 rpm



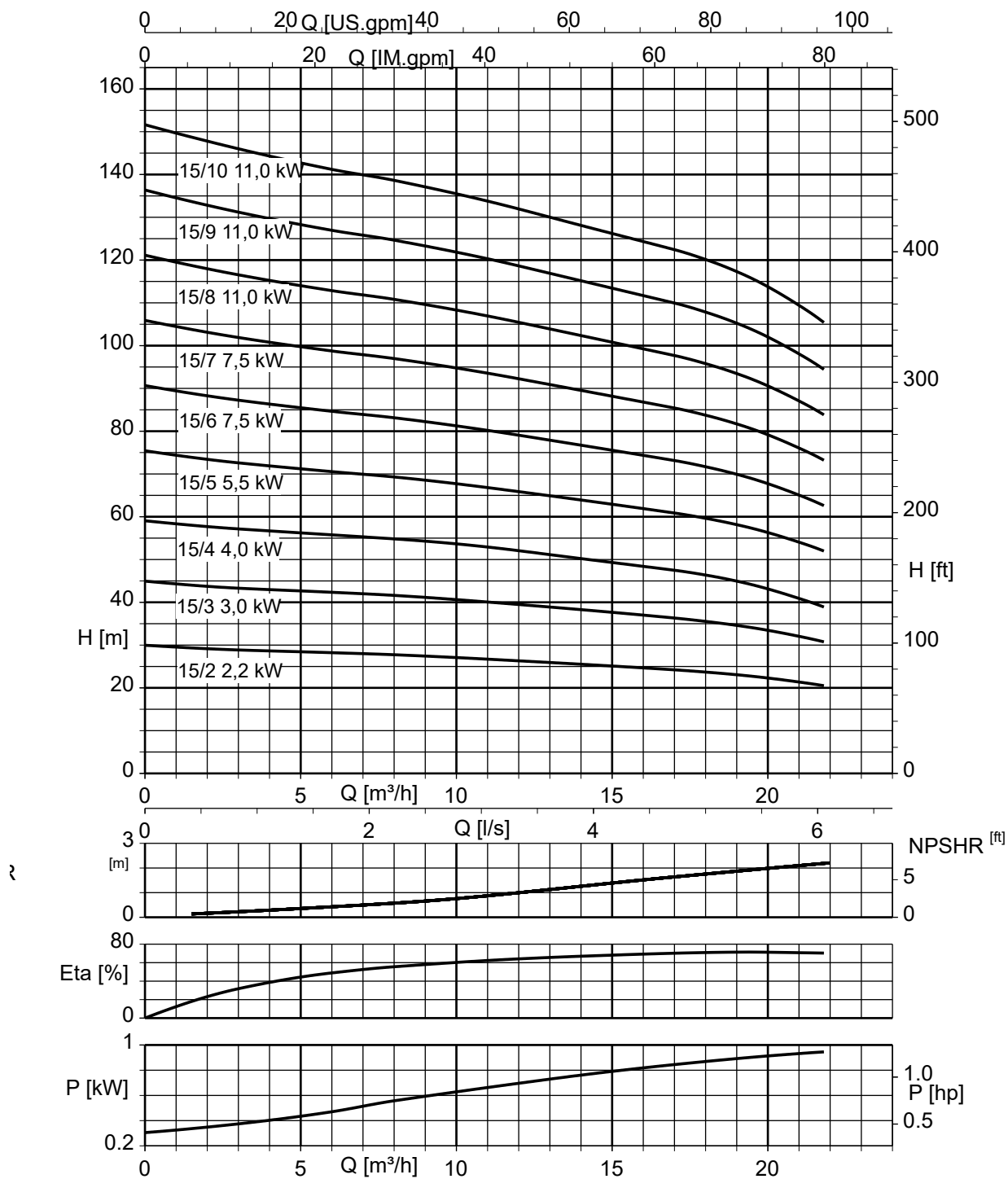
St = 1 | P per stage

HyaDuo 2 D FL with Movitec 10B, $n = 2900$ rpm



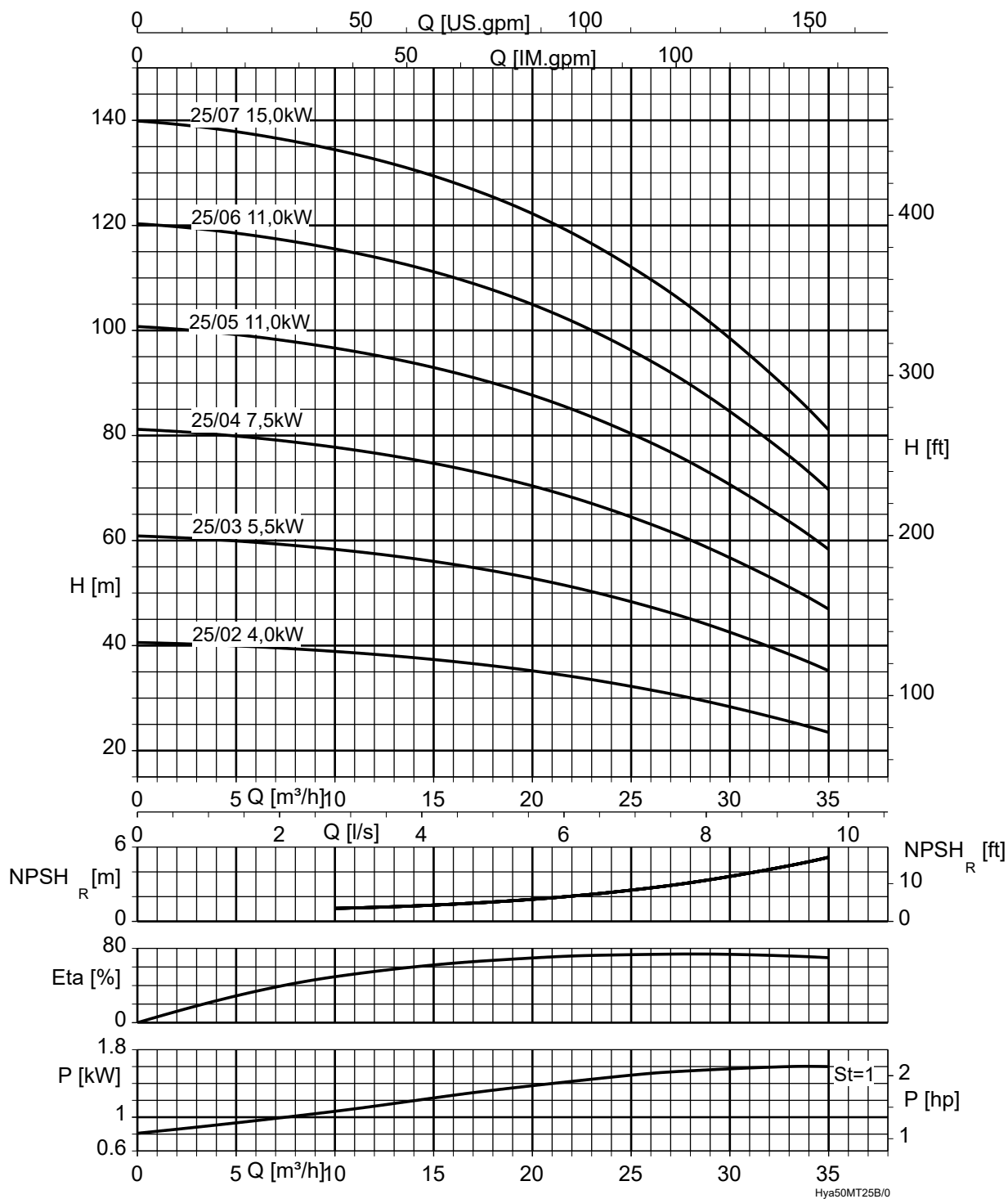
St = 1 | P per stage

HyaDuo 2 D FL with Movitec 15C, n = 2900 rpm



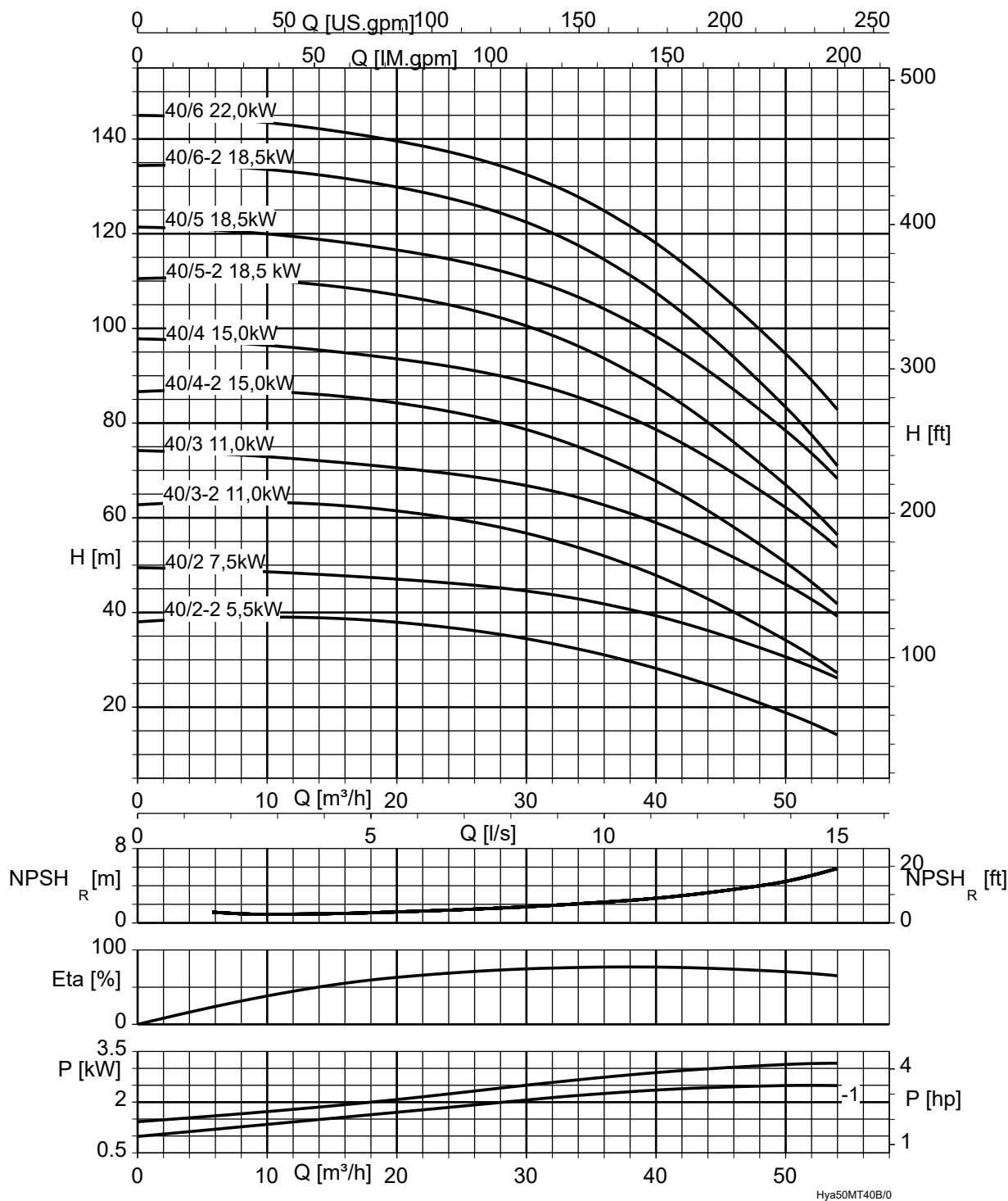
St = 1 | P per stage

HyaDuo 2 D FL with Movitec 25B, n = 2900 rpm



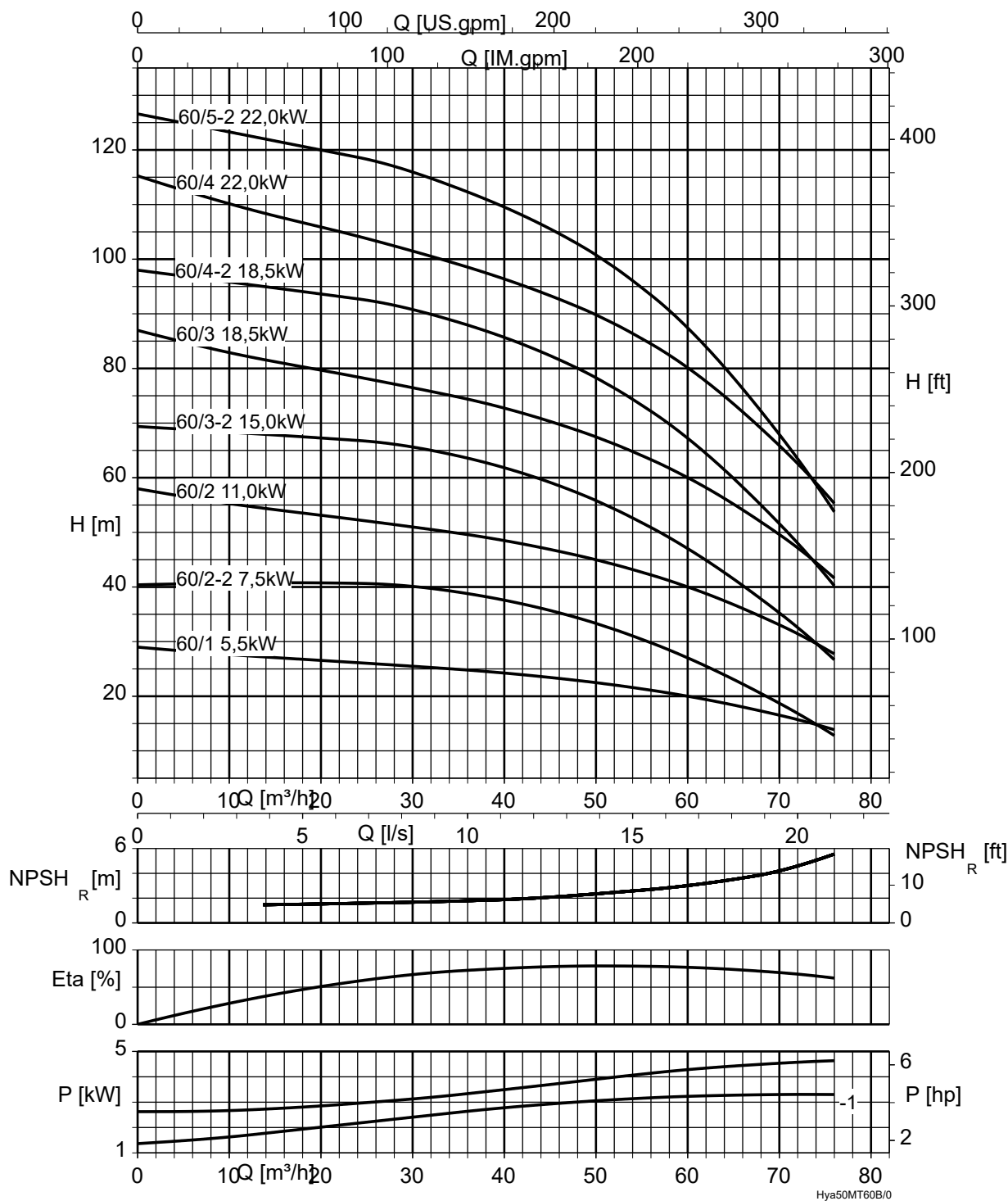
St = 1 | P per stage

HyaDuo 2 D FL with Movitec 40B, n = 2900 rpm



St = 1	P per stage	St = -1	P per stage with a smaller impeller
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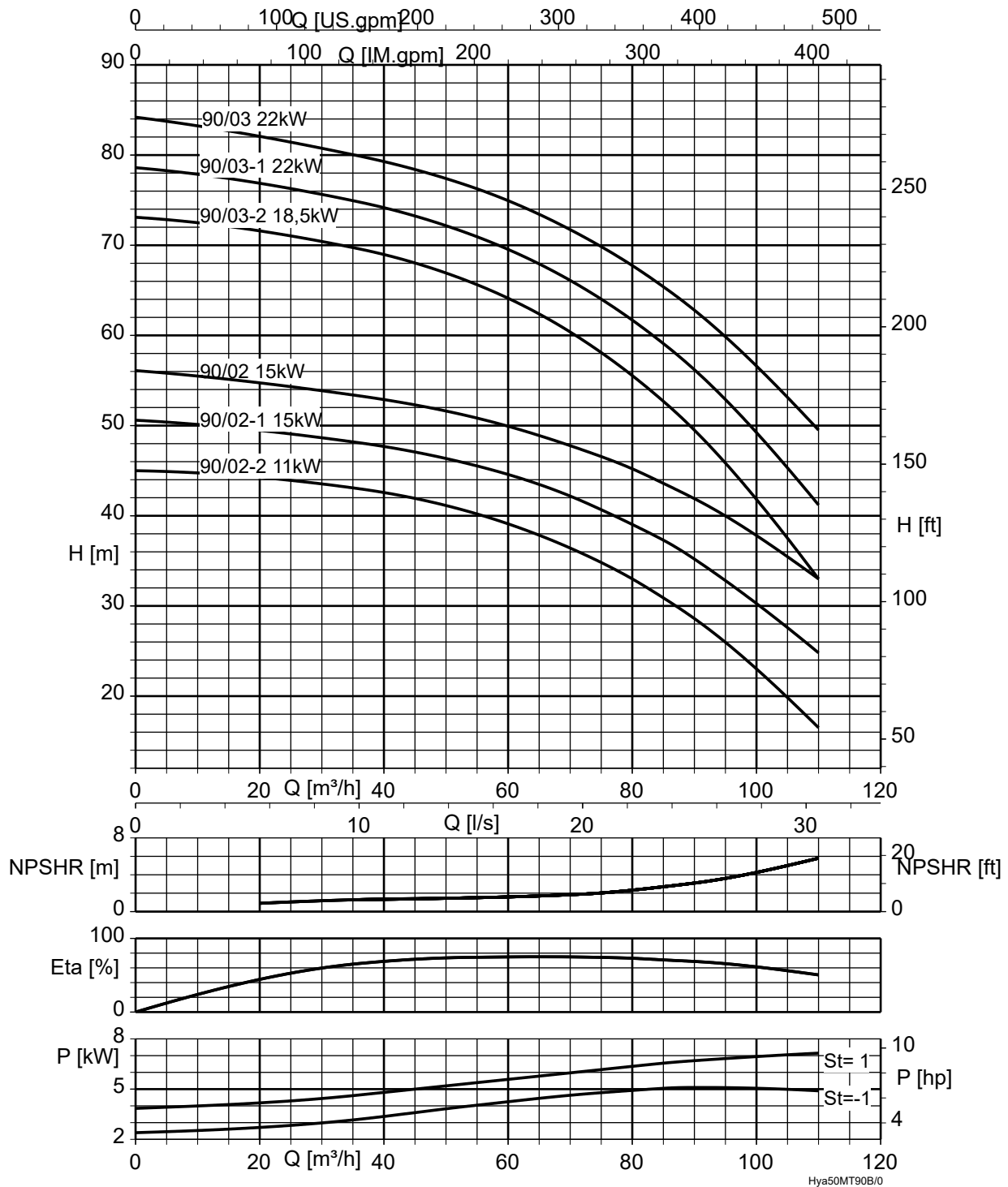
HyaDuo 2 D FL with Movitec 60B, $n = 2900$ rpm



St = 1 P per stage

St = -1 P per stage with a smaller impeller

HyaDuo 2 D FL with Movitec 90B, $n = 2900$ rpm



$St = 1$ | P per stage

$St = -1$ | P per stage with a smaller impeller

Dimensions

HyaDuo 2 D FL with Movitec 2B / 4B / 6B

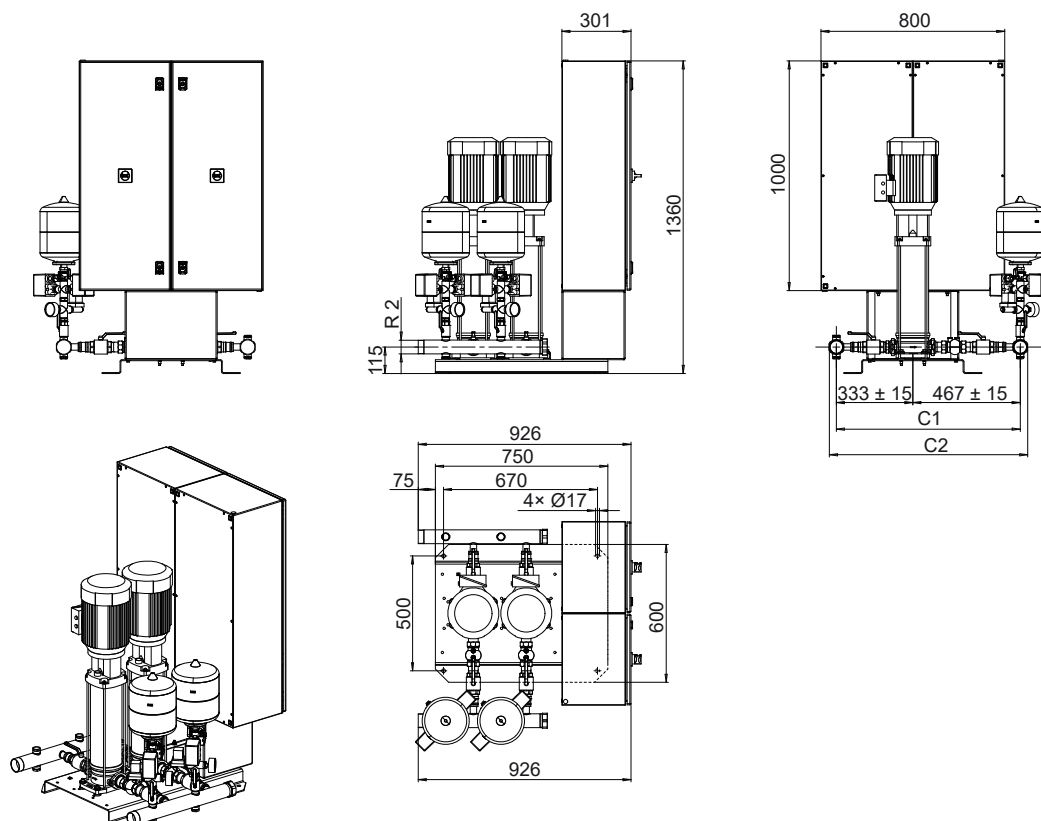


Fig. 2: Dimensions of HyaDuo 2 D FL with Movitec 2B / 4B / 6B [mm]

-  External thread R to DIN EN 10226
-  Flanges drilled to EN 1092-1 PN 16

Table 6: Dimensions

Size	C1	C2
	[mm]	[mm]
2 / 02...	742	802
2 / 04...	742	805
2 / 06...	800	863

HyaDuo 2 D FL with Movitec 10B

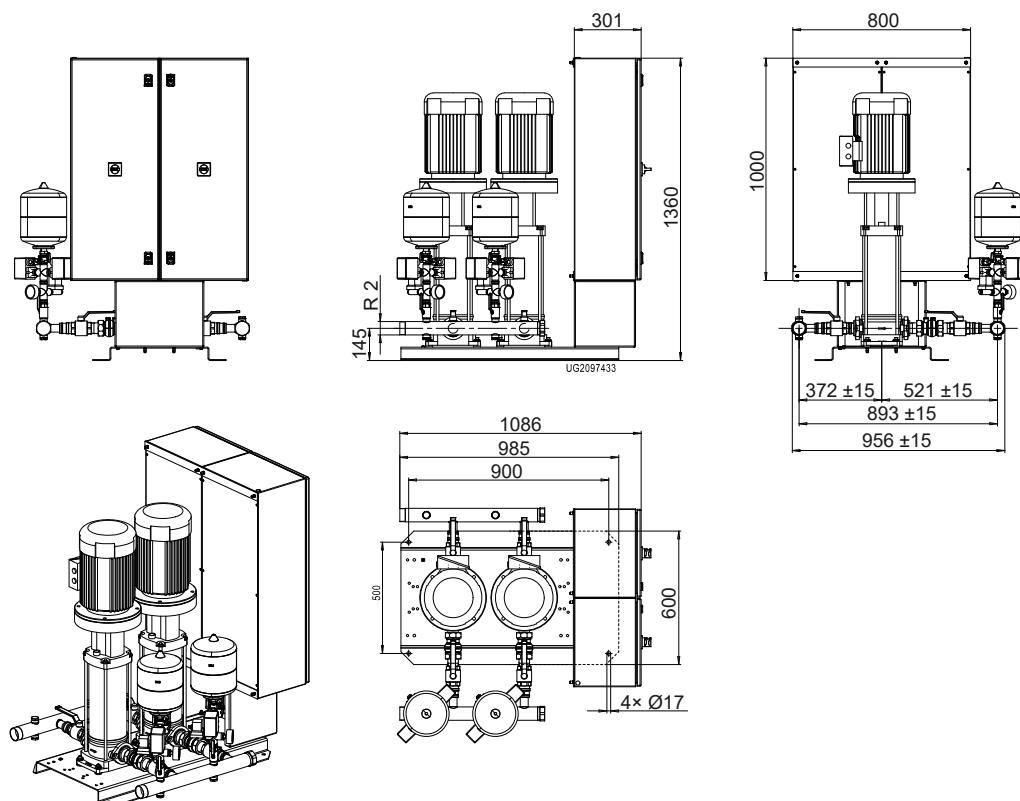


Fig. 3: Dimensions of HyaDuo 2 D FL with Movitec 10B [mm]

-  External thread R to DIN EN 10226
-  Flanges drilled to EN 1092-1 PN 16

HyaDuo 2 D FL with Movitec 15C

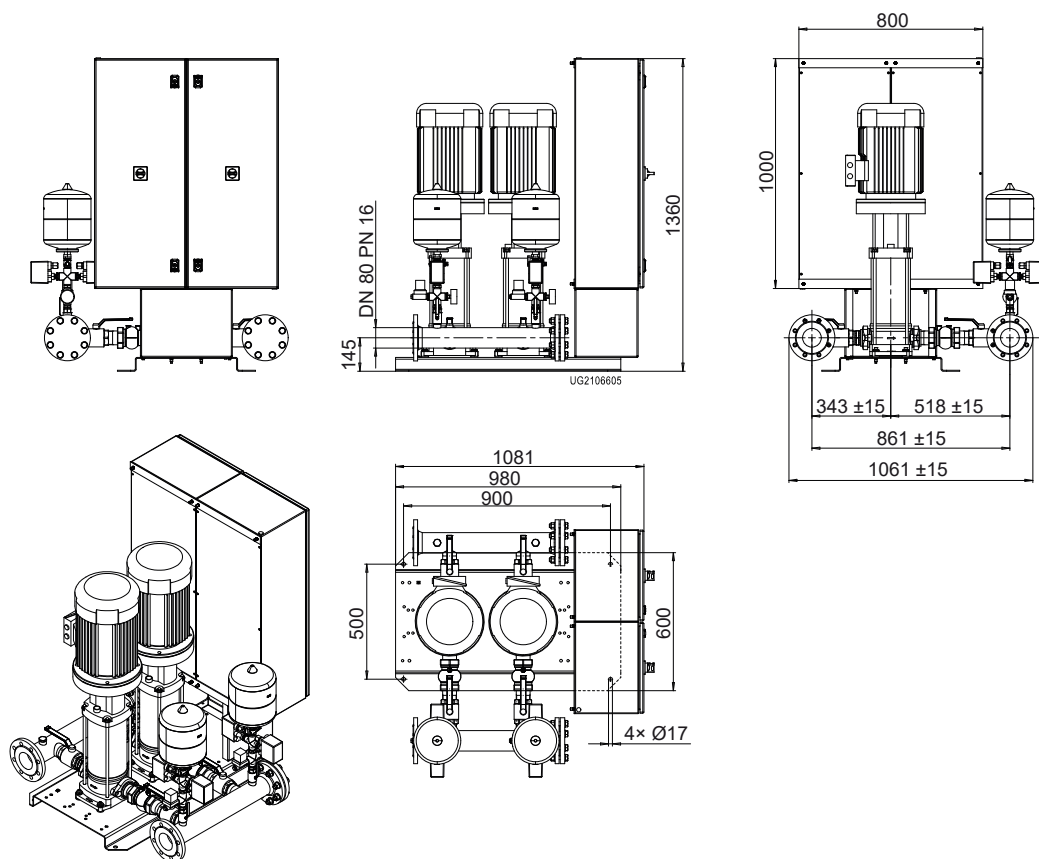


Fig. 4: Dimensions of HyaDuo 2 D FL with Movitec 15C [mm]

- i** External thread R to DIN EN 10226
- i** Flanges drilled to EN 1092-1 PN 16

HyaDuo 2 D FL with Movitec 25B / 40B / 60B / 90B

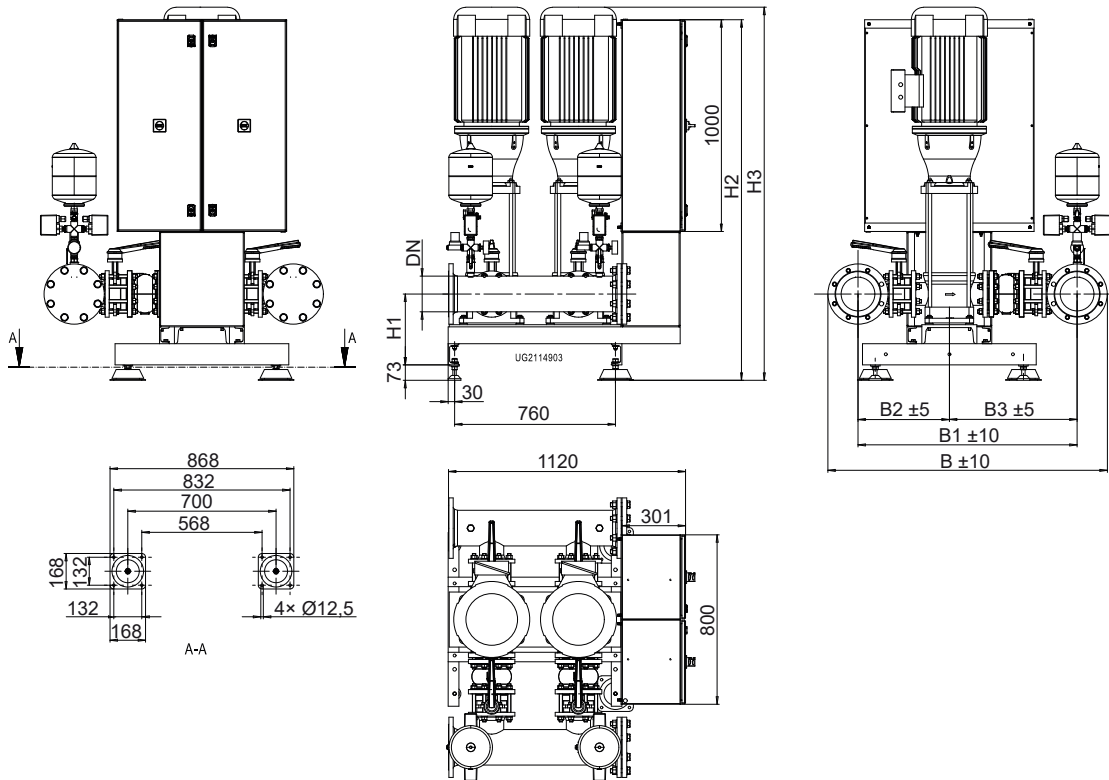


Fig. 5: Dimensions of HyaDuo 2 D FL with Movitec 25B / 40B / 60B / 90B [mm]

- External thread R to DIN EN 10226
- Flanges drilled to EN 1092-1 PN 16

Table 7: Dimensions

Size	DN	B	B1	B2	B3	H1	H2	H3
		[mm]	[mm]	[mm]	[mm]	[mm]	[mm]	[mm]
2 / 25...	100	1074	854	351	503	300	1553	1740
2 / 40...	100	1139	919	374	546	335	1553	1840
2 / 60...	150	1320	1035	431	604	335	1703	1762
2 / 90...	150	1335	1050	431	611	335	1703	1805

Scope of supply

Depending on the model, the following items are included in the scope of supply:

- Fire-fighting system
 - 2 vertical pump sets (Movitec high-pressure centrifugal pump incl. IE3 motor)
 - Base frame
 - Discharge-side piping
 - 2 check valves
 - 2 suction-side shut-off valves and 2 discharge-side shut-off valves
 - 2 discharge-side, direct-flow membrane-type accumulators, approved for drinking water
 - 2 pressure switches on the discharge side for starting and stopping, monitored for broken wires and short circuit
 - 2 pressure switches for monitoring pressure build-up
 - 2 pressure gauges
 - 2 temperature sensors
- Control unit
 - 2 power supply connections
 - 2 motor protection switches
 - 2 external master switches
 - 2 relays for broken wire monitoring / short-circuit monitoring
 - 2 manual-automatic selector switches
 - 2 timers
 - 2 phase monitoring relays
 - 2 soft starters from and including 5.5 kW
 - 2 control transformers
 - Terminal strip with markings for all connections

Accessories

 See the separate type series booklet Accessories for Pressure Booster Systems 1954.5.



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